

Cryptography-Encryption and Decryption

Aim

To perform Exploratory Data Analysis (EDA) in Python to understand the structure, relationships, and key patterns present in the dataset.

Procedure

1. Import the required libraries such as pandas, numpy, matplotlib.pyplot, and seaborn for data manipulation and visualization.
2. Load the dataset into a pandas DataFrame.
3. Display the first few rows using head() to get an initial overview of the data.
4. Use functions like info() and describe() to examine data types, non-null counts, and statistical summaries.
5. Check for missing values using isnull().sum() and decide whether to fill or drop them based on data context.
6. Identify duplicate records and remove them if necessary to maintain data quality.
7. Analyze the distribution of individual features using histograms or sns.displot() to understand spread and skewness.
8. Use boxplots to detect outliers in numerical features.
9. Explore relationships between variables using scatter plots, pair plots, or correlation heatmaps.
10. Summarize insights from the analysis such as trends, correlations, and anomalies that will guide further model building.

Program:

```
from cryptography.fernet import Fernet  
  
key = Fernet.generate_key()  
  
print("Generated Key:", key.decode())  
  
cipher = Fernet(key)  
  
data = "Sensitive Data"  
  
print("\nOriginal Data:", data)  
  
encrypted_data = cipher.encrypt(data.encode())  
  
print("Encrypted Data:", encrypted_data.decode())  
  
decrypted_data = cipher.decrypt(encrypted_data).decode()  
  
print("Decrypted Data:", decrypted_data)
```

Output:

Generated Key: d7dN0jF7ttiT5tKc0lfQYWH-
wUZ65E5xHeFpf3RUJsw=

Original Data: Sensitive Data

Encrypted Data: gAAAAABomVrGjUWv-ZL-M-
2mGTM9yTQgmutRWVA87CsAGG5GPhLh4cap8s8Mo_RQ69Oaz8S
4lCGZ4L60mEBY8rVHaApjBGZqMg==

Decrypted Data: Sensitive Data

Result:

Hence, a Python program to encrypt and decrypt sensitive data using the Fernet module from the Cryptography library has been implemented successfully, ensuring secure data transmission and confidentiality.